



DEVELOPMENT OF AN INTELLIGENT PROTOTYPE FOR PERCEPTION-ORIENTED RISK-AWARE DECISION-MAKING AND SPATIAL NAVIGATION IN ADAPTIVE CONFINED ENVIRONMENTS USING REINFORCEMENT LEARNING

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CHAPTER 1

Chapter 1 highlights the need to integrate human perception, emotion, and behavior into spatial path planning. It highlights existing limitations in current navigation systems, outlines the research problem, objectives, questions, and proposes the Theory of Planned Behavior (TPB) as the underpinning framework. The chapter also introduces the significance of applying AI to model perception-based navigation and sets the foundation for the conceptual and theoretical direction of the study.

CHAPTER 1

1.1 INTRODUCTION

According to Howard Gardner, human potential is expressed through nine distinct types of intelligence, each representing a unique way of perceiving, understanding, and interacting with the world. In today's rapidly evolving digital landscape, individuals are encouraged to cultivate and sharpen these intelligences to adapt and thrive. The need to digitise and augment these human capabilities has become increasingly important, paving the way for the development and widespread adoption of artificial intelligence. By integrating AI with multiple intelligences, we aim to amplify human potential and ensure technology meaningfully enhances our lives and societies



LINGUISTIC

Sensitivity to language, words, and rhetoric.

LOGICAL-MATHEMATICAL

Analytical problem-solving and logical reasoning.

MUSICAL

Skill in rhythm, pitch, and sound patterns.

BODILY KINESTHETIC

Control of body movements and object manipulation.

SPATIAL

Mental visualization and spatial reasoning.

INTERPERSONAL

Understanding social interactions and emotions.

INTRAPERSONAL

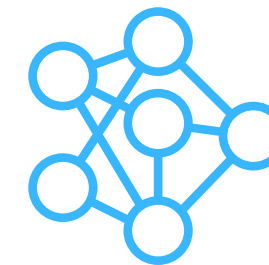
Self-awareness and introspective reflection.

NATURALISTIC

Sensitivity to natural elements and ecosystems.

EXISTENTIAL

Philosophical reflection on human existence/purpose.



DIGITALISATION
THROUGH
AI

VALUE ADDED THROUGH AI

NLP tools for real-time multilingual communication.

(Deyi Li et al, 2024)

AI algorithms optimize complex decision-making.

(Jian Wei et al, 2024)

AI-generated adaptive soundscapes for navigation cues.

(D. Ahmetovic et al, 2023)

Robotics/AI-guided precision in dynamic environments

(Y. Tong et al, 2024)

AI-driven 3D mapping (LiDAR-SLAM) for pathfinding

(J. Jung et al, 2024)

Emotion recognition AI for crowd-behavior prediction.

(H. -D. Le et al, 2023)

AI-driven personalized route suggestions (e.g., stress-aware paths).

(M. H. Abidi et al, 2024)

AI models simulating environmental adaptability.

(M. Reichstein et al, 2023)

Ethical AI frameworks for human-centric navigation.

(S. Cassini et al, 2025)

CHAPTER 1

1.2 BACKGROUND OF THE STUDY

Human Behaviour Quantification

The quantification of human behavior has significantly advanced through smartphone sensing methods, which utilize data from personal digital devices to measure behavior and physiology in real-time. This approach offers insights into health and behavior, transforming subjective observations into measurable data (Harari et al., 2017).

AI-Assisted Decision Making

Artificial intelligence enhances decision-making by rapidly analyzing large-scale behavioral data, identifying patterns, and reducing cognitive overload in complex environments. AI systems empower individuals and organizations with data-driven insights, improving accuracy and efficiency in various critical decisions, including healthcare, urban planning, and emergency response (Green & Chen, 2019; Buçinca et al., 2021).

Predictive AI for Human Behaviour

Predictive analytics powered by AI and behavioral modeling allow for improved forecasting of human movements and interactions. For example, AI systems have been developed to predict individual behavior in strategic games like chess, demonstrating the potential of AI in modeling and anticipating human actions (McIlroy-Young et al., 2020).

AI in Measuring Human Perception

AI now measures human perception using multimodal data such as EEG and eye-tracking, converting subjective perceptions into actionable insights. This capability provides unprecedented precision in assessing psychological comfort, stress levels, and emotional responses, greatly influencing the development of interactive technology and public safety measures (Katsigiannis & Ramzan, 2018).

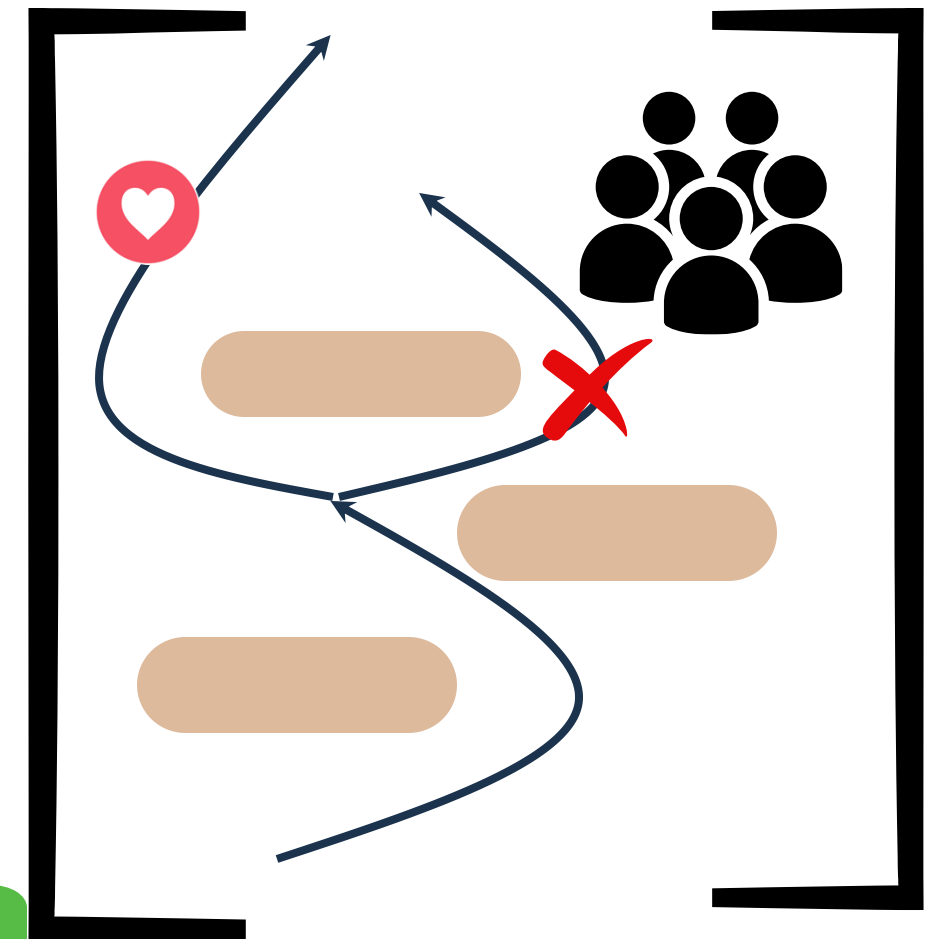
CHAPTER 1

1.3 PROBLEM STATEMENT

- Current pathfinding systems, such as Waze and Google Maps, primarily focus on shortest path algorithms and route frequency to optimize navigation. While Waze dynamically adjusts routes based on traffic density, it can result in inconvenient or less desirable paths such as narrow residential roads or complex intersections, which may affect user satisfaction (Leshed et al., 2008).
- On the other hand, Google Maps typically offers optimal routes based on efficiency but neglects the influence of human perception, such as comfort, safety, and stress triggers in spatial navigation (Montello, 2009).
- Additionally, navigation in confined spaces, such as malls, transit hubs, or event venues, often relies heavily on Simultaneous Localization and Mapping (SLAM), particularly through LiDAR technology, which emphasizes obstacle avoidance and 3D mapping rather than human-centric path optimization (Chen & Zhang, 2018).

This study addresses the gap by proposing a perception-based path forecasting and planning model, integrating cognitive and affective aspects of human decision-making into navigation systems. By incorporating perceptual dimensions such as environmental comfort, lighting, crowd density, and spatial openness, this model aims to enhance navigational experience and foster a more comfortable atmosphere in human movement through space (Zhang et al., 2018; Koelstra et al., 2018).

This human-aware approach will contribute to developing adaptive, emotion-sensitive AI systems capable of aligning route selection with users' psychological and physiological needs.



**Current
State**

Spatial Recognition

Pathfinding

Crowd
Behaviour

Perception-
based Decision
Making
(Forecast)

Crowd Control
(Management)

**Desired
State**



CHAPTER 1

1.4 RESEARCH QUESTION AND OBJECTIVE

Q u e s t i o n

RQ1

How can an AI-based framework be developed to enhance real-time spatial mapping and movement forecasting in confined environments ?

RQ2

What are the key factors influencing pedestrian route decision-making in confined spaces, particularly when occlusion, density fluctuations, and environmental constraints are present ?

RQ3

How do crowd density and herd dynamics influence pedestrian movement, and how can AI-driven models optimize flow efficiency in confined environments ?

RQ4

In what ways can AI models differentiate between the shortest path and the most preferred route based on human perception scores, stress levels, and environmental familiarity ?

RQ5

How can an interactive AI-powered spatial manipulation tool contribute to dynamic route optimization and pedestrian flow management?

RQ6

What key elements should be integrated into an AI-driven multimodal navigation model to improve spatial adaptability, human perception, and behavioral insights for real-world applications ?

R01

To develop an AI-based framework for real-time spatial mapping and movement forecasting in confined environments.

R02

To investigate the role of occlusion-aware AI models in enhancing pedestrian movement prediction and route decision-making.

R03

To analyze the influence of crowd density and herd dynamics on pedestrian movement within confined environments.

R04

To assess perception-based navigation preferences by analyzing human cognitive and emotional responses to spatial conditions.

R05

To design interactive AI-powered spatial manipulation tool for dynamic route optimization.

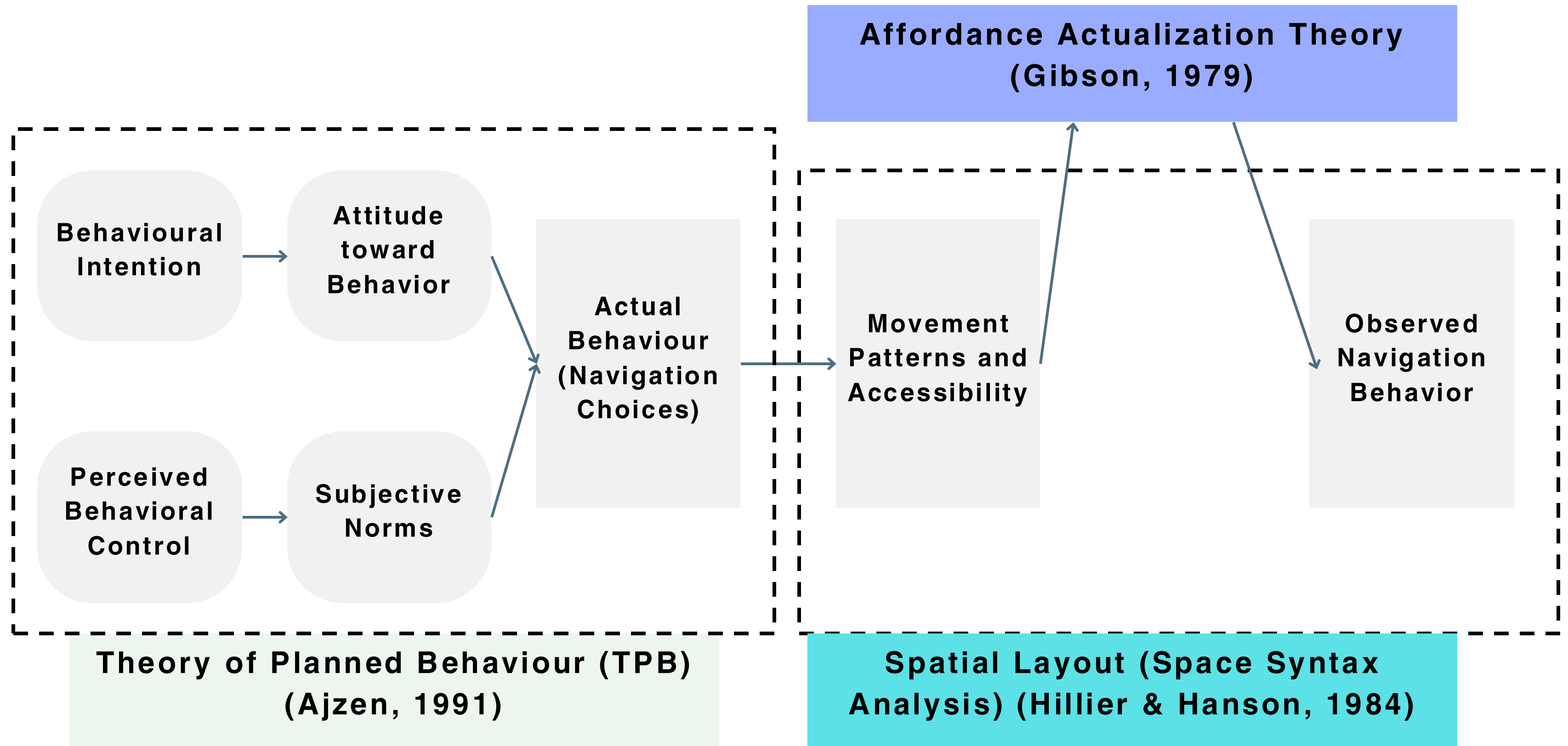
R05

To propose an AI-driven multimodal navigation model that integrates spatial adaptability, human perception, and behavioral insights for real-world implementation

O b j e c t i v e

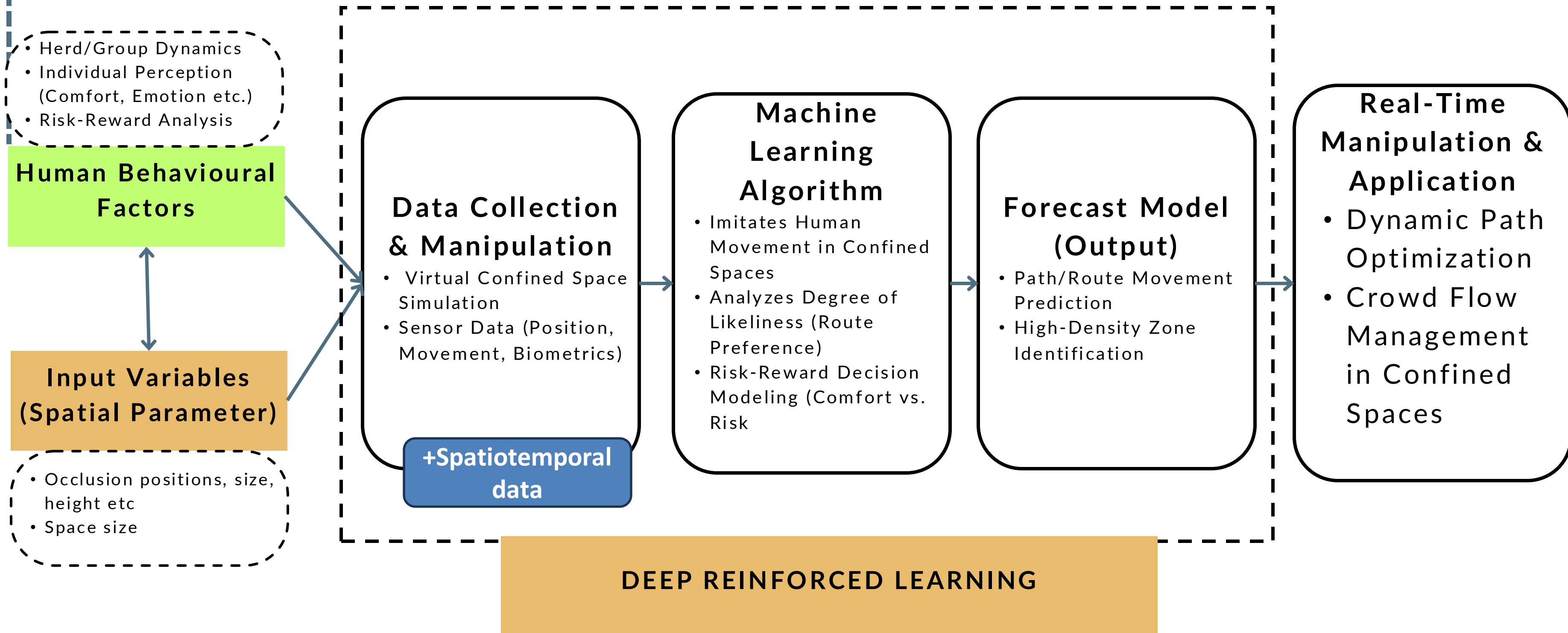
CHAPTER 1

1.5 THEORETICAL FRAMEWORK (UNDERPINNING THEORIES)



CHAPTER 1

1.6 CONCEPTUAL FRAMEWORK



CHAPTER 1

1.7 SIGNIFICANCE OF STUDY

Smart Spacing and Urbanisation

- Enhances urban space planning by optimizing pedestrian flows, improving comfort and spatial efficiency in urban layouts. (Chen et al., 2020)

Event Management

- Enables effective management of crowd density and route optimization during major events, enhancing attendee comfort and safety. (Wirz et al., 2022)

Development & City Planning
(Society/Economy)

Economic Efficiency &
Public Safety

Perception-Based Path Forecasting and Planning Algorithm (PB-PFP)

Emergency Response

- Supports rapid evacuation and emergency path planning by predicting crowd behavior under stress, guiding safer exit strategies and improving evacuation efficiency. (Lovreglio et al., 2019)

Public Safety & Disaster
Management

National Security &
Military Operations

Defense Applications

- Assists in strategic planning by forecasting pedestrian movements, optimizing navigation paths, and managing human flow in military or security-sensitive environments.
- Supports rapid response and adaptive evacuation strategies amid evolving security threats, including dynamic risks from advanced enemy technologies and aerial incursions. (Sharma & Ghosh, 2021)

CHAPTER 1

1.8 TERMINOLOGY AND DEFINITION

Perception-based Path Forecasting

Conceptual: A method to predict pedestrian routes considering human psychological perceptions, emotional responses, and environmental preferences (Wang et al., 2023).

Operational: Measured by algorithm-generated predictive accuracy of chosen paths relative to real-world observed pedestrian behaviour.

Virtual Confined Space

Conceptual: Digitally simulated environments used to replicate real-world confined areas to study human behavior under controlled scenarios (Nguyen et al., 2022).

Operational: Environment created using VR technology, measured by realism and accuracy of user biometric and positional responses in simulated scenarios.

Crowd Flow Management

Conceptual: Strategies implemented to regulate pedestrian movement effectively, preventing congestion, enhancing comfort, and ensuring safety in densely populated environments (Wirz et al., 2022).

Operational: Quantified by efficiency indicators, such as pedestrian throughput rates, evacuation times, and reduction of congestion-related incidents.



CHAPTER 2



Chapter 2 presents a comprehensive review of existing literature relevant to the human perception based decision making. It explores theories and concepts related to human perception, behavioral decision-making, path planning algorithms, crowd dynamics, and the integration of AI technologies.

CHAPTER 2

2.1 RELEVANT LITERATURE

	LITERATURE REVIEW	RESEARCH GAP	IMPLICATIONS OF THIS STUDY
Perception-based analytics	Studies on human behavior in navigation extensively focus on efficiency and physical metrics (shortest paths, quickest routes) (Sarker, 2021; Harari et al., 2017).	Limited research incorporating human perception and emotional states into navigational decision-making.	This study emphasizes integrating perceptual factors like comfort and emotional responses into navigation algorithms, improving user experience beyond efficiency metrics alone.
Human Wellbeing	Urban and spatial planning traditionally prioritizes practical factors such as accessibility and infrastructure rather than ambiance and emotional well-being (Chen et al., 2020).	Insufficient focus on creating atmospheres that affect mood, comfort, and psychological well-being in pedestrian pathways.	The study addresses this by developing a framework that explicitly considers atmospheric factors (lighting, spatial openness) to enhance psychological comfort in urban navigation.
Technology Limitation	Machine learning and virtual reality have advanced simulations and predictive analytics, their integration into real-time human behavior and path prediction remains limited due to computational complexity and the need for efficient algorithms (González et al., 2016; Al Machot et al., 2022).	The limited application of real-time predictive analytics in human behavior and path prediction presents a gap in developing responsive systems.	This research aims to develop computationally efficient, real-time predictive models that integrate human behavior analysis, enhancing the responsiveness and adaptability of navigation systems.



THANK YOU

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